

Ruhila Goswami

✉ ruhila@ieee.org
🌐 RuhiRG



“Nothing exists for itself alone, but only in relation to other forms of life.”
– Charles Darwin

Personal Data

Name Ruhila Goswami
Date Of Birth 20.09.2001

Education

2024–2026 **M.S. Biological Science**, *University of Iceland, Reyjavik, Iceland*
(ADVISOR: Prof. Arnar Pálsson)
(THESIS: Evolutionary Divergence in Dental Variation and Sex Chromosome Architecture in Icelandic Brown Trout and Arctic Charr)
2021–2024 **B.S. Biology (Major)**, *Indian Institute of Science Education and Research (IISER), Mohali, India*
First Division with Honors
2020 **Intermediate (AISSCE)**, *Velammal Vidyalaya, Ayanabakkam, Chennai, India*
83.8% Central Board of Secondary Education (CBSE)
2017 **High School (AISSE)**, *Velammal Vidyalaya, Ayanabakkam, Chennai, India*
10.0 Cumulative Grade Point Average (CGPA) in Central Board of Secondary Education (CBSE)

Experience

Internships

- SUMMER 2023 **Prof. Arnar Pálsson**, *University of Iceland*, Research Staff
Studied the syntenic of Histone genes for Salmonids, focusing on their arrangement and neighboring genomic regions.
- WINTER 2022 **Prof. Debabrata Goswami**, *IIT Kanpur*, Research Intern
I familiarized myself the basics of non-linear optics and its usage in optical tweezers for cessation of cancerous cells and microscopy.
- WINTER 2022 **Dr. Nagma Parveen**, *IIT Kanpur*, Research Intern
I familiarized myself with wet-lab methods for studying the mechanisms of viral action under varying external stimuli. I also began work on a protocol for the generation of pseudoviruses for SARS-CoV-2 based on an HIV-1 lentiviral packaging system with a luciferase reporter.
- SUMMER 2022 **Prof. Arnar Pálsson**, *University of Iceland*, Research Staff
Detailed analysis in a literate and reproducible manner for simulating a series of possible molecular evolutionary pathways for the *Salmolidae* using phylogenetic trees. This involved the five steps on an HPC (High Performance Computer) with literate programming visualization in R:
 - Data curation with NCBI databases
 - Homology inference using similarity measures (BLAST)
 - Multiple sequence alignment (MUSCLE5)
 - Alignment trimming (G-BLOCK)
 - Tree simulation with distance measures (BIONJ) and maximum likelihood approaches (RAXML-NG)

PROJECT REPORT: Computational Primitives for Evolutionary Paths ($\approx$ 147 pages)
- SUMMER 2021 **Dr. Lolitika Mandal**, *IISER Mohali*, Research Intern
Exploring Genetic Tools for working with *Drosophila* from a wet-lab perspective of data collection and analysis.
- Volunteer Work
- JUNE-AUGUST 2024 **pydSEAMSlib: Enhanced Python bindings for d-SEAMS**, *Google Summer of Code*, Student Developer
This was continuation of my last year's project. This project focused on improving the usability and accessibility of 'pySEAMS' by creating comprehensive documentation, refining the installation process, and integrating new features to make the tool more user-friendly
- JUNE-AUGUST 2023 **PYSEAMS: Python bindings for d-SEAMS**, *Google Summer of Code*, Student Developer
Developed Python bindings for d-SEAMS, a molecular dynamics analysis engine, enabling qualitative ice structure classification in confined and bulk systems. Refactored C++ code, removed lua bindings, and enhanced usability and functionality for MacOS users. Contributed to the seams-core engine and pyseams repository, improving documentation, automated testing, and code management. Gained skills in compiled languages, binding creation, open-source collaboration, and project management.
- 2022–PRESENT **IEEE P3173**, *IEEE Standards Committee*, Secretary
Am supporting the drafting the IEEE Standard for Endocrine Disrupting Chemical Hazard Labeling

Teaching Experience

- SPRING 2026 **LÍF401G Developmental Biology**, *University of Iceland*, Teaching Assistant
Course led by Prof. Adam Ray Smith. 8 credit undergraduate course on developmental processes across vertebrates and invertebrates, from organismal development to molecular regulation, with practical work in microscopy, embryo analysis, and developmental genetics databases.
- FALL 2025 **LÍF541G Plant Physiology**, *University of Iceland*, Teaching Assistant
Course led by Prof. Victor Anthony Albert. 7.5 credit undergraduate course on flowering plant physiology, including growth, hormone and environmental responses, nutrient transport, carbon and nitrogen metabolism, plant-microbe interactions, reproduction, and laboratory exercises.
- SPRING 2025 **LÍF403G Evolutionary Biology**, *University of Iceland*, Teaching Assistant
8 credit undergraduate course on evolutionary theory, natural selection, adaptation, and the history of life.

Certifications

NPTEL Courses

- APRIL 2024 **Biophotonics**, *IIT Kharagpur*, Distinction, 87%
- APRIL 2024 **Introduction to Professional Scientific communication**, *IIT Kanpur*, 79%
- SEP 2022 **Applications of machine learning techniques in biology using WEKA**, *IIT Madras*, Distinction, 93%

Other Courses

- NOV 2022 **Practical Python for beginners: a biochemist's guide**, *Biochemical Society, U.K.*

Technical Skills

Programming Languages

EXPERIENCED	R, Python (3.x), Shell (zsh,bash)	FAMILIAR	C, Java
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Bioinformatics Packages

EXPERIENCED	Randomized Maximum Likelihood new generation (RAXML-NG), MUSCLE5 (multiple sequence alignment)	FAMILIAR	WEKA, BEAST2 (Bayesian Evolutionary Analysis Sampling Trees) via babette, Snakemake
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Tools

EXPERIENCED Git (version control), ssh, Vim, Markdown

FAMILIAR Office-Suites (MS, OpenOffice, LibreOffice)

Experimental

FLY Handling flies (wild-type, w¹¹¹⁸, tubby), Drosophila larva dissection (brain, salivary gland, proventriculus, imaginal discs, gastric caeca), Fixing, staining, mounting viewing tissues with Fluorescent microscopes, Making PBS, PFA

FISH Dissection, otolith extraction, bone extraction, DNA extraction and preservation, DNA sex determination via PCR, DNA sequencing, electro-fishing

CELLS Cell culture

PROFESSIONAL Time management, critical thinking, problem solving, communication

Research Topics

EXPERIENCED Phylogenetic Tree Construction (Distance, Maximum Likelihood, Bayesian), Evolutionary Biology, Population genetics, R reproducible literate programming, High performance open source software, Scientific Software Development for High Throughput calculations, synteny

INTERESTED Population genetics, phenotypic-genotypic mapping, structural variation in genomes and selection, salmonids, non-invasive methods

Affiliations

Memberships

2024–PRESENT **Nordic Society Oikos**, Student Member

2024–PRESENT **Icelandic Ecological Society (Vistis)**, Student Member

2024–PRESENT **The Icelandic Biological Society (Líffræðigáttin)**, Student Member

2022–PRESENT **RSC (Royal Society of Chemistry)**, Student Member

2022–PRESENT **RSB (Royal Society of Biology)**, Student Member

2022–PRESENT **British Ecological Society**, Student Member

2022–PRESENT **Biochemical Society, UK**, Undergraduate Member
 2022–PRESENT **Genetics Society, UK**, Undergraduate Member
 2022–PRESENT **Genetics Society of America**, Undergraduate Member
 2022–PRESENT **Royal Microscopical Society, UK**, Undergraduate Member
 2022–PRESENT **IEEE EMBS (Engineering in Medicine and Biology Society)**, Student Member
 2022–PRESENT **Federation of European Biochemical Societies (FEBS)**, Member
 2022–PRESENT **European Microscopy Society**, Member
 2022–PRESENT **ACM (Association for Computing Machinery)**, Student Member

Publications

MANUSCRIPTS IN PREPARATION

IN Evolutionary divergence in tooth numbers and bone shape by populations and
 PREPARATION species in Icelandic salmonids

CONFERENCE PROCEEDINGS

Rohit Goswami, Ruhila S, Amrita Goswami, Sonaly Goswami, and Debabrata Goswami. “Reproducible High Performance Computing Without Redundancy with Nix.” In: *2022 Seventh International Conference on Parallel, Distributed and Grid Computing (PDGC)*. 2023.

Rohit Goswami and Ruhila S. “High Throughput Reproducible Literate Phylogenetic Analysis (accepted).” In: *2022 Seventh International Conference on Parallel, Distributed and Grid Computing (PDGC)*. 2023.

PREPRINTS

Rohit Goswami, Ruhila S, Amrita Goswami, Sonaly Goswami, and Debabrata Goswami. *Unified Software Design Patterns for Simulated Annealing*. Feb. 6, 2023. arXiv: 2302.02811 [physics]. URL: <http://arxiv.org/abs/2302.02811> (visited on 02/10/2023).

Conference Records

Posters

- APRIL 2026 **Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids**, *Icelandic Ecological Society (Vistís)*, Ruhila S.
- MARCH 2026 **Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids**, *NOWPAS*, Ruhila S.
- OCTOBER 2025 **Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids**, *IceBio*, Ruhila S.

- 2025 **Diversity in craniofacial elements by populations and sexes in salmonids of Iceland**, *Icelandic Ecological Society (Vistís)*, Ruhila S., Best Poster Award
- NOVEMBER 2022 **Tracing Lineages of *Salmo Salar* through Histone sequence data**, *BES Annual Meeting 2022*, Ruhila S.
- Oral Presentations
- NOVEMBER 2022 **High Throughput Reproducible Literate Phylogenetic Analysis**, *IEEE PDGC 2022*, R. Goswami, Ruhila S.
- NOVEMBER 2022 **Reproducible High Performance Computing Without Redundancy with Nix**, *IEEE PDGC 2022*, R. Goswami, Ruhila S., A. Goswami, S. Goswami and D. Goswami
- NOVEMBER 2022 **Reproducible Literate Programming Workflows for Censored Data**, *IOP Machine Learning for Healthcare*, R. Goswami, R. S.